Shanmukha Ramakrishna Vedantam

AI Leader — Senior Director, Iris Labs (Flagship Pioneering)

Cambridge, MA • Green Card Holder

LinkedIn • Google Scholar • +1-650-407-7845 • vrama@gatech.edu

Executive Summary

AI leader and scientist building humanlike AI. My research began with understanding human learning patterns and actuating them with AI — teaching machines to learn concepts, generalise, and imagine as people do. It now runs in the opposite direction: using AI to influence and empower human neural networks, most directly in iris chat's emotional support conversations. My work at this intersection of human and AI systems has produced 20+ papers and 60K+ citations, and helped launch a B2C product in the AI + wellness space.

Work Experience

Senior Director Nov. 2025 - Present

Iris Labs (Flagship Pioneering), Cambridge, MA

Research leadership at Iris Labs, Flagship Pioneering's company for transforming human wellbeing through AI, whose first product is iris chat, an AI thought partner for emotional support. My role is enabling others to be effective: I build the research prototypes, evaluation systems, and analysis tooling that research, product, and engineering teams draw on, and validate approaches that others then integrate. Most of this work sits upstream of the shipped product — the pieces currently live in the iris chat system are the LLM judges and the iris tags.

- **Self-Improvement and Human Feedback:** Led the build-out of LLM judges for validation and safety, improved from human feedback — the piece of this research currently live in the iris chat system; it anchors benchmarking and self-improvement for emotional support quality
- **Agentic Platform:** Created **Agent Control Center**, my own agentic orchestration platform (a custom Claude Code–style system) integrating RLM implementations and context-tagging techniques — used to analyze and deliver insights for agentic product-manager efforts and to prototype custom, high-quality onboarding flows for others to take forward; also built **Katana**, a research prototype that reads a user's stance and emotional state and maps them to actual responses
- **AI-Driven Product Feedback:** Built **Iris Palantir**, a privacy-aware system to analyze, tag, and aggregate conversations across the platform with PII considerations built in — turning aggregate conversation statistics into product improvements and go-to-market (GTM) decisions; the iris tags it produces are also live in the iris chat system today
- **Hiring:** Make strategic hiring decisions for technical candidates in machine learning — including hiring 1 FTE (Ph.D. in Computer Science, UW–Madison) to the research team

AI Evangelist, Angel Investor, and Researcher Jan. 2024 - Nov. 2025

Self-Employed, New York, NY

- **Democratizing AI Training:** Pushed dataset distillation as a route to capable models at a fraction of the training cost — *Embarrassingly Simple Dataset Distillation* (ICLR 2024) — and co-organized the first Dataset Distillation Challenge (ECCV 2024) to build the research community around it
- **Safety and Last-Mile Deployment:** Bridged frontier research to broader AI-safety narratives, and backed deployment-stage companies as an angel investor (Artificio AI, Sphere.app)

Visiting Faculty Feb. 2023 - Dec. 2023

NYU Center for Data Science

- **Data-Efficient and Robust Multimodal AI:** Led a research programme across data-efficient learning, multimodal representation, and distributionally robust AI — including the dataset-distillation work that produced *Embarrassingly Simple Dataset Distillation* (ICLR 2024), selective visual question answering (CVPR 2023), and the class-level effects of data augmentation (NeurIPS 2023)

Research Scientist Feb. 2019 - Feb. 2023

Meta Inc., Fundamental AI Research (FAIR), NYC

Formulated, owned and drove an ambitious research agenda on building provably robust AI that shares similar cognitive underpinnings as humans

- **Humanlike Concept Learning:** Pursued how machines can learn productive, compositional concepts the way people do — CURI, a benchmark for productive concept learning under uncertainty (ICML 2021), and probabilistic neural-symbolic models for interpretable visual question answering (ICML 2019)
- **Foundations of Robust Generalization:** Characterised when learned representations generalise out-of-distribution — the Decodable Information Bottleneck (NeurIPS 2020 Spotlight), an empirical investigation of domain generalization (NeurIPS 2021), and the nullspace account of OOD failure (ICLR 2023)
- **Structured Multimodal Representations:** Gave image–text models the hierarchical geometry their concepts actually have — MERU hyperbolic representations (ICML 2023; Best Paper, ICLR MRL Workshop) and COAT, measuring compositionality in emergent representations (ICML 2022)
- **Mentorship:** Mentored interns and AI Residents who have since gone to Google DeepMind, Meta AI, WorldLabs, and Stanford; released code, datasets, and resources for the AI community across these projects

Education

- **Georgia Tech, United States** 2013–2018
  Ph.D, Computer Science (transfer from Virginia Tech)
  Thesis: Interpretation, Grounding and Imagination for Machine Intelligence
  Committee: Devi Parikh (Advisor), Larry Zitnick, Kevin P. Murphy, Jacob Eisenstein, Dhruv Batra
- **Virginia Tech, United States** 2013–2016
  M.S., Computer Engineering
  Advisor: Prof. Devi Parikh
  Specialization: Computer Vision and Machine Learning
- **International Institute of Information Technology (IIIT), Hyderabad, India** 2009–2013
  Bachelor of Technology, Electronics and Communication Engg.
  Advisor: Prof. K. Madhava Krishna
  Specialization: Vision for Robotics

Honors and Achievements

Research Accomplishments and Awards

1. 60,906 citations with an H-index of 19 Oct. 2026
2. Best Paper Award, ICLR workshop on Multimodal Representation Learning (MRL) 2023
3. Google PhD Fellowship in Machine Perception, Speech Technology and Computer Vision
   One out of 5 awardees selected across North America, Europe and the Middle East 2018
4. ICLR travel award for attending the International Conference on Learning Representations 2018
5. Finalist for the Adobe Research Fellowship 2018
6. Finalist for the Adobe Research Fellowship 2016
7. Awarded travel grant of USD 1000 for CVPR, 2017 under Google's Archimedes program 2017

Reviewing

1. Outstanding reviewer award at ICLR 2021

2. Outstanding reviewer award at ICCV 2019

3. Outstanding reviewer award at CVPR — Awarded to 130 reviewers in the CVPR reviewer pool 2017

Conference Publications[1]

1. **Emphasizing Discriminative Features for Dataset Distillation in Complex Scenarios.**
Kai Wang, Zhe Li, Zhi-Qiang Cheng, Samira Khaki, Ahmad Sajedi, Ramakrishna Vedantam, Konstantinos N. Plataniotis, et al.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [31 citations]

2. **Embarrassingly Simple Dataset Distillation.**
Yunzhen Feng, Ramakrishna Vedantam, Julia Kempe.
International Conference on Learning Representations (ICLR), 2024 [45 citations]

3. **Understanding the detrimental class-level effects of data augmentation.**
Polina Kirichenko, Mark Ibrahim, Randall Balestriero, Diane Bouchacourt, Ramakrishna Vedantam, Hamed Firooz, Andrew Gordon Wilson.
Neural Information Processing Systems (NeurIPS), 2023 [36 citations]

4. **Hyperbolic Image-Text Representations.**
Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, Ramakrishna Vedantam.
International Conference on Machine Learning (ICML), 2023 [296 citations]
Best Paper Award, Multimodal Representation Learning (MRL) Workshop colocated with ICLR, 2023

5. **Improving Selective Visual Question Answering by Learning from Your Peers.**
Corentin Dancette, Spencer Whitehead, Rishabh Maheshwary, Ramakrishna Vedantam, Stefan Scherer, Xinlei Chen, Matthieu Cord, Marcus Rohrbach.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [47 citations]

6. **Dont forget the nullspace! Nullspace occupancy as a mechanism for out of distribution failure.**
Daksh Idnani, Vivek Madan, Naman Goyal, David J. Schwab, Ramakrishna Vedantam.
International Conference on Learning Representations (ICLR), 2023 [8 citations]

7. **COAT: Measuring Object Compositionality in Emergent Representations.**
Sirui Xie, Ari S Morcos, Song-Chun Zhu, Ramakrishna Vedantam.
International Conference on Machine Learning (ICML), 2022 (Short Oral) [10 citations]

8. **An Empirical Investigation of Domain Generalization in Empirical Risk Minimizers.**
Ramakrishna Vedantam, David Lopez-Paz, David J. Schwab.
Neural Information Processing Systems (NeurIPS), 2021 [76 citations]

9. **CURI: A Benchmark for Productive Concept Learning Under Uncertainty.**
Ramakrishna Vedantam, Arthur Szlam, Maximilian Nickel, Ari Morcos, Brenden Lake.
International Conference on Machine Learning (ICML), 2021 (Short Oral) [37 citations]

10. **Learning Optimal Representations with the Decodable Information Bottleneck.**
Yann Dubois, Douwe Kiela, David J Schwab, Ramakrishna Vedantam.
Neural Information Processing Systems (NeurIPS), 2020 (Spotlight) [Top 4%] [74 citations]

11. **IR-VIC: Unsupervised Discovery of Sub-goals for Transfer in RL.**
Nirbhay Modhe, Prithvijit Chattopadhyay, Mohit Sharma, Abhishek Das, Devi Parikh, Dhruv Batra, Ramakrishna Vedantam.
International Joint Conference on Artificial Intelligence (IJCAI), 2020 [Top 12.6%] [6 citations]
Also presented at *ICLR Workshop on Task Agnostic Reinforcement Learning, 2019*

12. **Probabilistic Neural-Symbolic Models for Interpretable Visual Question Answering.**
Ramakrishna Vedantam, Karan Desai, Stefan Lee, Marcus Rohrbach, Dhruv Batra, Devi Parikh.
International Conference on Machine Learning (ICML), 2019 (Long Oral) [Top 4.2%] [125 citations]

[1] Publications in top AI conferences (CVPR, ICML, NeurIPS, ICCV) typically have 20-25% acceptance rates. Citation counts from Google Scholar, Oct. 2026.

13. **Generative Models of Visually Grounded Imagination.**
Ramakrishna Vedantam, Ian Fischer, Jonathan Huang, Kevin P. Murphy.
International Conference on Learning Representations (ICLR), 2018 [Top 10%] [196 citations]

14. **Grad-CAM: Why did you say that? Visual Explanations from Deep Networks via Gradient-based Localization.**
Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra.
International Conference on Computer Vision (ICCV), 2017 [47,563 citations]
Also presented at *NIPS Workshop on Interpretable Machine Learning in Complex Systems, 2016*

15. **Sound-Word2Vec: Learning Word Representations Grounded in Sounds.**
Ashwin K. Vijayakumar, Ramakrishna Vedantam, Devi Parikh.
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017 [37 citations]

16. **Counting Everyday Objects in Everyday Scenes.**
Prithvijit Chattopadhyay, Ramakrishna Vedantam, Ramprasaath R. Selvaraju, Dhruv Batra, Devi Parikh.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017 (Spotlight) [Top 8.2%] [235 citations]

17. **Context-aware Captions from Context-agnostic Supervision.**
Ramakrishna Vedantam, Samy Bengio, Kevin P. Murphy, Devi Parikh, Gal Chechik.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017 (Spotlight) [Top 8.2%] [177 citations]
Also presented as an Oral at the *Bay Area Machine Learning Symposium (BayLearn), 2017*

18. **Visual Word2Vec (vis-w2v): Learning Visually Grounded Word Embeddings using Abstract Scenes.**
Satwik Kottur, Ramakrishna Vedantam, Jose Moura, and Devi Parikh.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016 [124 citations]

19. **Learning Common Sense through Visual Abstraction.**
Ramakrishna Vedantam, Xiao Lin, Tanmay Batra, C. Lawrence Zitnick, and Devi Parikh.
IEEE International Conference on Computer Vision (ICCV), 2015 [114 citations]
Also presented as an oral at *1st Workshop on Object Understanding for Interaction*, colocated with *ICCV, 2015*

20. **CIDEr: Consensus-based Image Description Evaluation.**
Ramakrishna Vedantam, C. Lawrence Zitnick, and Devi Parikh.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2015 [7,714 citations] *Equal Contribution*

Journal Publications

1. **Adopting Abstract Images for Semantic Scene Understanding.**
C. Lawrence Zitnick, Ramakrishna Vedantam, and Devi Parikh.
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 2016 [107 citations]
Special Issue on the best papers at the 2013 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

2. **Grad-CAM: Why did you say that? Visual Explanations from Deep Networks via Gradient-based Localization.**
Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra.
International Journal of Computer Vision (IJCV), 2020

Workshops and arXiv Manuscripts

- **Embarrassingly Simple Dataset Distillation.** Yunzhen Feng, Ramakrishna Vedantam, Julia Kempe. *Workshop on Advancing Neural Network Training (WANT), NeurIPS, 2023*
- **Hyperbolic Image-Text Representations.** Karan Desai, Maximilian Nickel, Justin Johnson, Ramakrishna Vedantam. *ICLR Workshop on Multimodal Representation Learning (MRL), 2023 (Spotlight) [Best Paper Award]*
- **Understanding the class-specific effects of data augmentations.** Polina Kirichenko, Randall Balestriero, Mark Ibrahim, Ramakrishna Vedantam, Hamed Firooz, Andrew Gordon Wilson. *ICLR Workshop on Trustworthy ML, 2023*
- **Microsoft COCO Captions: Data Collection and Evaluation Server.** Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Dollar, C. Lawrence Zitnick. *arXiv:1504.00325* [3,848 citations]

Professional Services

Area Chair: NeurIPS 2026; British Machine Vision Conference (BMVC), 2022
Chair: Technical Chair, Workshop on Applications of Deep Learning (WALDA) 3.0, 2023
Workshop Organization: The First Dataset Distillation Challenge, co-located with ECCV 2024
Reviewing:
Conference: Reviewer for ICCV 2015-2019, 2025; CVPR 2016-2020, 2023-24, 2025; ECCV 2016-2020; ACCV 2016; BMVC 2017; NeurIPS 2017-2020, 2022; ICLR 2018-2022; ICML 2018-2019; UAI 2018-2021, 2025
Journals: Reviewer for International Journal of Computer Vision (IJCV), IEEE Transactions on Image Processing, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Neural Computation, Computer Speech and Language, IEEE Open Journal of Signal Processing (2022)
Panelist on the "All Things Attention" Workshop 2022 (co-located with NeurIPS, 2022). Volunteer for Women in Machine Learning (WiML) mentoring program, 2022. Volunteered in organizing Mid-Atlantic Computer Vision (MACV) workshop at Virginia Tech.

Past Research Internships

Research Internship Summer 2018
Microsoft Research Cambridge, United Kingdom
Generative Models for Concept Learning
Supervisor: Nate Kushman, Matthew Johnson, and Sebastian Nowozin, Microsoft Research

Research Internship Summer 2017
Facebook AI Research (FAIR), Menlo Park, CA
Supervisor: Devi Parikh, Dhruv Batra and Marcus Rohrbach, Facebook AI Research/Georgia Tech

Research Internship Winter 2017
Machine Perception Group, Google Research, Mountain View, CA
Grounded latent variable generative models for images and semantics.
Supervisor: Kevin P. Murphy, and Ian Fischer, Google Research

Research Internship Summer 2016
Machine Perception Group, Google Research, Mountain View, CA
Worked on a system to explain class discrimination conditioned on an image, using natural language. Given an image, a target category and a distractor, explain why the image contains the target.
Supervisor: Gal Chechik, and Samy Bengio, Google Research and Google Brain

Research Internship Summer 2014
Center for Visual Computing, Ecole Centrale de Paris/ INRIA-Saclay, France
Worked on Loopy Part Models for Face Detection
Supervisor: Prof. Iasonas Kokkinos, Ecole Centrale de Paris

Internship and AI Resident Advising

- Yunzhen Feng (Co-advised with Julia Kempe), New York University (NYU) — Summer-Fall 2023. Current Position: Ph.D. student at NYU
- Polina Kirichenko (NYU-FAIR collaboration), Facebook AI Research — Fall 2022. Current Position: Research Scientist at Meta AI
- Karan Desai (Summer Intern), Facebook AI Research — Summer 2022. Current position: Founding Member of Technical Staff at WorldLabs Inc.
- Daksh Idnani (AI Resident), Facebook AI Research — Fall 2021 - Spring 2022. Current position: Founder and CTO, Sphere.app
- Sirui Xie (Summer Intern), Facebook AI Research — Summer 2021. Current position: Research Scientist at Google DeepMind
- Yann Dubois (AI Resident), Facebook AI Research — Fall 2019 - Spring 2020. Current position: Ph.D. student at Stanford University
- Siddharth Ancha (Summer Intern), Facebook AI Research — Summer 2019. Current position: Postdoc at Massachusetts Institute of Technology (MIT)
- Prithvijit Chattopadhyay (Summer Intern), Virginia Tech — Summer 2015. Current position: Research Scientist at NVIDIA Research
- Satwik Kottur (Winter Intern), Virginia Tech — Fall 2014. Current position: Research Scientist at Meta AI

Thesis Co-Supervision

Georgia Institute of Technology (Gatech)
Ananya Raval (Co-advised with Devi Parikh) — Generation of Linux Commands Using Natural Language Descriptions (M.S. Thesis), May 2018

Open Source Contributions

Lead developer of the Domainbed Measures project opensourced by Facebook research
Lead developer of the Productive Concept Learning project opensourced by Facebook research
Lead developer of the Joint VAE project open sourced by Google
Developer on the coco-caption project which implements commonly used image captioning metrics such as CIDEr, METEOR, BLEU, and ROUGE-L.
Developer of the CIDEr project which implements the two versions of CIDEr (CIDEr and CIDEr-D) from our CVPR'15 paper on Consensus Based Image Description Evaluation.

Talks

- **Tutorial on Multimodal Machine Learning** — NYU Center for Data Science (Andrew Gordon Wilson Lab), March 2023
- **Measuring and Structuring Generalizable Deep Representations** — NYU Center for Data Science, Feb. 2023; Department of Computer Science, Princeton University, Feb. 2023; Indian Institute of Science (IISc), May 2023; Indian Institute of Technology (IIT), Delhi, April 2023; Indian Institute of Technology (IIT), Madras, April 2023; International Institute of Information Technology (IIIT), Hyderabad, May 2023; Indian Institute of Technology (IIT), Hyderabad, April 2023; Department of Computer Science, University of Michigan, Sep. 2023; Stevens Institute of Technology, Feb. 2024
- **Concept Abstraction and Generalization for Machine Learning** — ConCats Seminar, New York University, Oct. 2020; Open Data Science Conference (ODSC), Boston, March 2022; AAAI Symposium on Conceptual Abstraction and Analogy, Nov. 2020
- **Interpretation, Grounding, and Imagination** — Indian Institute of Technology (IIT), Kanpur, May 2021; Courant Institute, New York University; University of Oxford; Google DeepMind, London; Microsoft Research, Cambridge; Facebook AI Research, Menlo Park; University of California, Berkeley; Google, Mountain View; Allen Institute for AI Research (AI2), Seattle; Toyota Technological Institute (TTI), Chicago; Indian Institute of Science (IISc), Bangalore; International Institute of Information Technology (IIIT), Hyderabad (2017–2019)

- **Connecting Vision and Language** — via invited talks at the institutions above
- Panel Discussion — "All things Attention" Workshop 2022 (co-located with NeurIPS, 2022)

Coursework

Graduate Coursework: Computer Vision Systems, Advanced Computer Vision, Introduction to Machine Learning, Probabilistic Graphical Models, Independent Study-Deep Learning, Numerical Analysis and Software, Data Analytics-2, Convex Optimization, Deep Learning for Perception, Bayesian Statistics, Mathematical Foundations of Machine Learning, Computability and Algorithms
Selected Undergraduate Coursework: Mobile Robotics, Artificial Neural Networks, Speech Signal Processing, Medical Image Processing, Engineering Systems, Data Structures, Operating Systems and Algorithms

Skills

Programming Languages: Python, C++, Lua, Matlab
Libraries: PyTorch, TensorFlow, Caffe, NLTK (Natural Language ToolKit)
Human Computation: Amazon Mechanical Turk

Extra Curricular

Hosted all the Talks at Felicity - 2011, annual college fest of IIIT Hyderabad
Coordinator and Founder - Entrepreneurship Cell at IIIT Hyderabad
Class Representative for ECE Undergraduate batch
Member, Students Parliament (Monsoon 2012 and Spring 2013)
Campus Ambassador for Teach for India at IIIT (2011 to 2012)
Trained in Carnatic Classical music for 7 years
Regular participation in Computer Vision and Machine Learning Reading Group at Virginia Tech